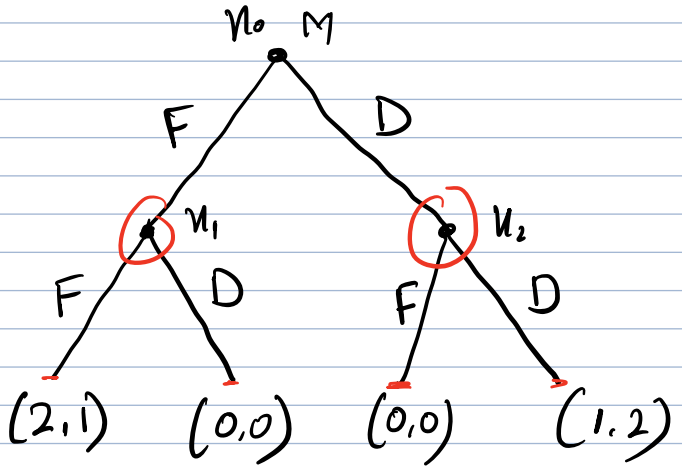
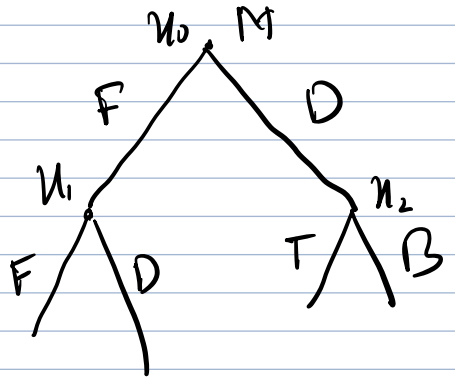
Class 16



(F, F), (F, D)
(D, F), (D, D)



A direct rooted tree

A direct rooted tree is a tripled $T = (V, E, n_0)$

where

- V is a finite set with $|V| \geq 2$. The elements are called noted / vertices.
- $E \subseteq V \times V$. The elements are called edges. An edge $(x, y) \in E$ is an outgoing edge of x and an incoming edge of y.
- $n_0 \in V$ is called the root of the tree. For every $x \in V$, there is a unique path from

x_0 to x where a path is a series of edges $(x_0, x_1), (x_1, x_2), \dots, (x_{k-1}, x)$ for some $k \geq 0$.

- x_0 has no incoming edges.

Defn
The nodes that do not have any outgoing edges are called end nodes or leaves. We denote the set of leaves by $Z (\subseteq V)$

Defn for a $x \in V$, $C(x)$ denotes the children of x, $C(x) = \{ y \in V \mid (x, y) \in E \}$

Defn for a node $x \in V$, $A(x) = \{ e \in E \mid e = (x, y) \text{ for some } y \in V \}$

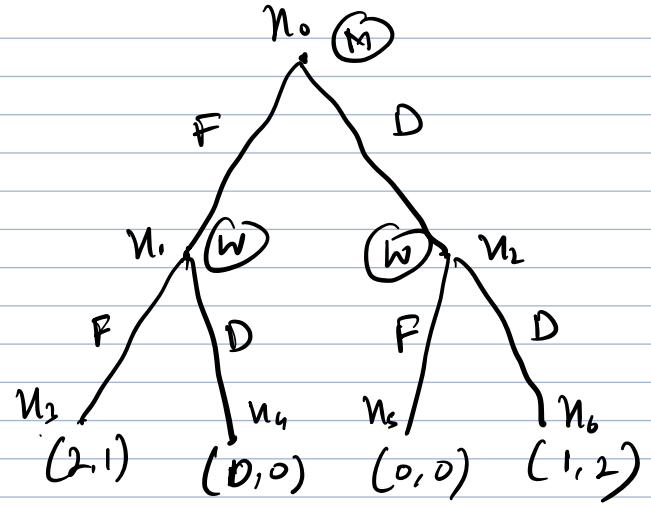
Defn of extensive form game.

An extensive form game is a tuple

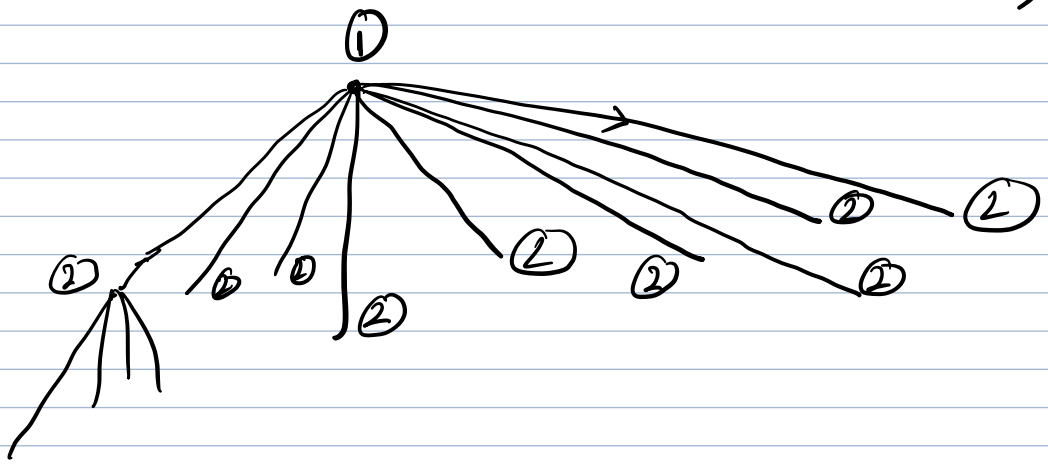
$$\Gamma = \langle N, V, E, x_0, (V_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

where

- N is a finite set of players.
- (V, E, x_0) is a directed rooted tree.
- $(V_i)_{i \in N}$ is a partition of the nodes that are not the leaves, i.e. of the set $V \setminus Z$.
- $(u_i)_{i \in N}$ is a collection of utility functions where $u_i : Z \to \mathbb{R}$ for all $i \in N$.



- $N = \{M, W\}$
- $V = \{n_0, n_1, n_2, n_3, n_4, n_5, n_6\}$
- $E = \{(n_0, n_1), (n_0, n_2), (n_1, n_3), (n_1, n_4), (n_2, n_5), (n_2, n_6)\}$
- $n_0 = n_0$
- $(V_i)_{i \in N}$: $V_M = \{n_0\}$, $V_W = \{n_1, n_2\}$
- $u_1(n_3) = 2$, $u_1(n_4) = u_1(n_5) = 0$, $u_1(n_6) = 1$
  $u_2(n_3) = 1$, $u_2(n_4) = u_2(n_5) = 0$, $u_2(n_6) = 2$



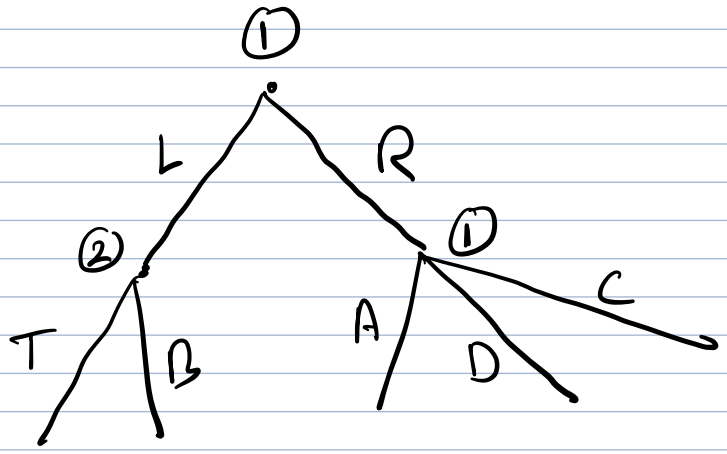
- Let $J : V \setminus Z \to N$ $\quad J(n) = i$ if $n \in V_i$

The game process in the following way

- Player $J(n_0)$ initiates the game by choosing a

possible actions from $A(n_0)$, Equivalently, he chooses a node $n_1 \in C(n_0)$.

- n_1 is a leaf, the game ends there. Other. $J(n_1)$ chooses a node $n_2 \in C(n_1)$.
- In this way, the game continues.

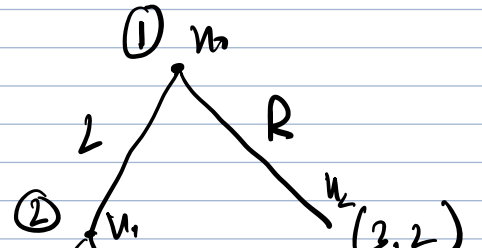


- These games are perfect information games. The players know what has happened be making their moves.

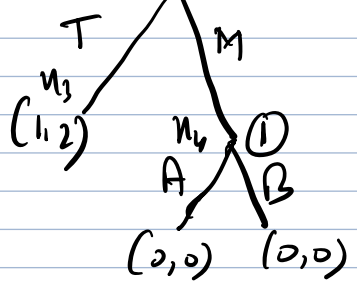
- A strategy for player i, s_i, is a mapping which maps each $n \in V_i$ to an element of $A(n)$ (equivalently to an element of $C(n)$)

$$s_i : V_i \longrightarrow \bigcup_{n \in V_i} A(n)$$

$$\text{s.t. } s_i(n) \in A(n) \quad \forall n \in V_i$$



$V_M = \{n_0\}$
$s_1(n_0) = R$
$\bar{s}_1(n_0) = D$



$s_2(n_4) = F$

$s_2(n_2) = F$

$V_1 = \{n_0, n_4\}$, $V_2 = \{n_1\}$

$s_1(n_0) = R$

$s_1(n_4) = A / B$